

Decimals:

While Addition / Subtraction,
there should be equal digits
after decimal.

Q1. $12.5 + 3.75 = ?$

- (A) 16.05 (B) 15.25 ☒ (C) 16.25 (D) 16.75 (E) 17.25

$$12.50 + 3.75$$

$$\begin{array}{r} 12.50 \\ + 3.75 \\ \hline 16.25 \end{array}$$

Q2. $0.008 + 0.09 + 1.2 = ?$

- ☒ (A) 1.298 (B) 1.28 (C) 1.208 (D) 1.098 (E) 1.398

$$0.008 + 0.090 + 1.200$$

$$= 1 + 0.298$$

$$= 1.298$$

Q3. $(-2.75) + 5.6 - 0.85 = ?$

- (A) 1 (B) 1.5 (C) 3 (D) 2.5 ☒ (E) 2

$$= -2.75 + 5.60 - 0.85$$

$$= 5.60 - 2.75 - 0.85$$

$$= 3.00 + (0.60 - 0.75 - 0.85)$$

$$= 3 + (-1.00) \Rightarrow 3 - 1 = 2$$

Q4. $(0.8 + 0.08 + 0.008) - 0.088 = ?$

- ☒ (A) 0.8 (B) 0.808 (C) 0.88 (D) 0.78 (E) 0.08

$$0.800 + 0.080 + 0.008 - 0.088$$

$$0.800 + (0.008) - 0.088$$

$$= 0.8$$

Q5. $0.4 + 0.44 + 0.444 + 0.4444 + 0.44444 = ?$

- (A) 2.17284 (B) 2.17248 (C) 2.7284 (D) 1.7284 (E) 2.22222

$$0.40000 + 0.44000 + 0.44400 + 0.44440 + 0.44444$$

$$= 2.17284 \quad \{ \text{Addition} \}$$

Q6. $6.66 + 6.066 + 66.666 + 66.060 + 0.066 = ?$

- (A) 145.518 (B) 145.458 (C) 138.858 (D) 140.518 (E) None of these

$$= 6.660 + 6.066 + 66.666 + 66.060 + 0.066$$

$$= 144 + (0.660 + 0.066 + 0.666 + 0.060 + 0.066)$$

$$= 144 + 1.518$$

$$= 145.518$$

Q7. $0.006 \times 0.03 = ?$

- (A) 0.00018 (B) 0.0018 (C) 0.000018 (D) 0.18 (E) 0.0006

$$= 0.006 \times 0.03$$

$$= \frac{6}{1000} \times \frac{3}{100} = \frac{18}{100000} = 0.00018$$

Q8. $0.125 \times 0.64 = ?$

- (A) 0.8 (B) 0.08 (C) 0.008 (D) 0.064 (E) 0.16

$$= 0.125 \times 0.64$$

$$= \left(\frac{125}{1000} \right) \times \left(\frac{64}{100} \right) \Rightarrow \frac{1}{8} \times \frac{64}{100} \Rightarrow \frac{8}{100} \Rightarrow 0.08$$

Also, $125 = \frac{1000}{8} \Rightarrow 12.5 = \frac{100}{8} \Rightarrow 1.25 = \frac{10}{8}$

and $0.125 = \frac{1}{8}$

So, in question,

$$0.125 \times 0.64 \Rightarrow \frac{1}{8} \times \frac{64}{100} = 0.08$$

Q9. $0.3 \times 0.07 \times 1000 = ?$

- (A) 2.1 (B) 210 (C) 21 (D) 0.21 (E) 0.021

$$\begin{aligned} &= 0.3 \times 0.07 \times 1000 \\ &= \frac{3}{10} \times \frac{7}{100} \times 1000 = 21 \end{aligned}$$

Q10. $1.6 \times 0.0625 = ?$

- (A) 0.16 (B) 0.01 (C) 0.1 (D) 1 (E) 0.0625

$$\begin{aligned} &= 1.6 \times 0.0625 \\ &= \frac{16}{10} \times \frac{625}{10000} \\ &= \frac{16}{10} \times \frac{1}{16} = \frac{1}{10} = 0.1 \end{aligned}$$

$$\left\{ \begin{aligned} 6.25 \div 100 &= \frac{1}{16} \\ 0.0625 &= \frac{1}{16} \end{aligned} \right\}$$

Q11. $7.2 \div 0.8 = ?$

- (A) 7 (B) 0.9 (C) 90 (D) 8 (E) 9

$$7.2 \div 0.8$$

$$\begin{aligned} \frac{72}{10} \div \frac{8}{10} &\Rightarrow \frac{72}{10} \times \frac{10}{8} \\ &= 9 \end{aligned}$$

{ Reciprocal }

Q12. $0.45 \div 0.09 = ?$

$$\# \frac{a}{b} \div \frac{c}{d} = \frac{a \times d}{b \times c}$$

$$\left\{ \begin{aligned} &\text{like,} \\ &\frac{5}{7} \div \frac{9}{8} = \frac{5 \times 8}{7 \times 9} \end{aligned} \right\}$$

$$* \frac{7/8}{3/7} = \frac{7 \times 7}{8 \times 3} = \frac{49}{24}$$

$$* \frac{4/5}{10/7} = \frac{4 \times 7}{5 \times 10} = \frac{28}{50}$$

(A) 50

(B) 0.5

☒ (C) 5

(D) 4

(E) 0.05

$$= 0.45 \div 0.09$$

$$= \frac{45}{100} \div \frac{9}{100} \Rightarrow \frac{45}{100} \times \frac{100}{9} \quad \{ \text{Reciprocal} \}$$

$$= 5$$

Q13. $0.0072 \div 0.0009 = ?$

(A) 9

(B) 0.8

(C) 80

(D) 8

(E) 7

$$= 0.0072 \div 0.0009$$

$$= \frac{72}{10000} \div \frac{9}{10000}$$

{ Reciprocal }

$$= \frac{72}{10000} \times \frac{10000}{9} = 8$$

Q14. $5.04 \div 0.6 = ?$ ☒ (A) 8.4

(B) 84

(C) 0.84

(D) 9.4

(E) 7.4

$$= 5.04 \div 0.6$$

$$= \frac{504}{100} \div \frac{6}{10} \Rightarrow \frac{504}{100} \times \frac{10}{6} = \frac{84}{10} \Rightarrow 8.4.$$

Q15. $\{ (3.6 \div 0.12) - (2.5 \times 0.8) \} = ?$

(A) 26

☒ (B) 28

(C) 30

(D) 32

(E) 24

$$= \left\{ \left(\frac{36}{10} \div \frac{12}{100} \right) - \left(\frac{25}{10} \times \frac{8}{10} \right) \right\}$$

$$= \left(\frac{36}{10} \times \frac{100}{12} \right) - \left(\frac{25}{10} \times \frac{8}{10} \right)$$

$$= 30 - 2$$

$$= 28$$

Q16. $(0.75 + 1.25) \times (0.6 \div 0.2) = ?$

- (A) 3 (B) 2 (C) 9 (D) 12 (E) 6

$$= (0.75 + 1.25) \times (0.6 \div 0.2)$$

$$= (2.00) \times \left(\frac{6}{10} \times \frac{10}{2} \right)$$

$$= 2 \times 3$$

$$= 6$$

Q17. $[(4.05 - 2.85) \div 0.3] + (0.04 \times 250) = ?$

- (A) 10 (B) 12 (C) 14 (D) 16 (E) 18

$$= \left[(4.05 - 2.85) \div \frac{3}{10} \right] + \left[\frac{4}{100} \times 250 \right]$$

$$= \left(1.20 \times \frac{10}{3} \right) + 10$$

$$= \left(\frac{120}{100} \times \frac{10}{3} \right) + 10 \Rightarrow 4 + 10 \Rightarrow 14$$

Q18. $100 - \{ (0.99 \div 0.11) + (0.12 \div 0.03) \} = ?$

- (A) 83 (B) 87 (C) 86 (D) 89 (E) 90

$$= 100 - \left\{ \left(\frac{99}{100} \times \frac{100}{11} \right) + \left(\frac{12}{100} \times \frac{100}{3} \right) \right\}$$

$$= 100 - (9 + 4) \Rightarrow 100 - 13 = 87$$

Q19. $(3/5 \text{ of } 2.5) + (0.4 \times 3.75) = ?$

- (A) 2.5 (B) 3 (C) 3.5 (D) 4 (E) 1.5

$$= \left(\frac{3}{5} \times \frac{5}{2} \right) + \left(\frac{4}{10} \times \frac{375}{100} \right)$$

$$= \left(\frac{3}{2} \right) + \left(\frac{4}{10} \times \frac{15}{4} \right)$$

$$= \frac{3}{2} + \frac{3}{2} \Rightarrow \frac{6}{2} = 3$$

$$\left\{ \begin{aligned} 3.75 &= 3 + \frac{3}{4} \\ &= \frac{15}{4} \end{aligned} \right\}$$

Q20. $(\sqrt{0.0004} \div 0.02) + (0.3 \times 0.3 \times 100) = ?$

(A) 9

(B) 12

(C) 11

☒ (D) 10

(E) 8

$$= \left\{ \sqrt{\frac{4}{10000}} \div \frac{2}{100} \right\} + \left(\frac{3}{10} \times \frac{3}{10} \times 100 \right)$$

$$= \left(\frac{2}{100} \times \frac{100}{2} \right) + (9)$$

$$= 1 + 9 = 10$$

Q21. $(0.125 \div 0.05) \times (0.4 + 0.6) = ?$

☒ (A) 2.5

(B) 25

(C) 0.25

(D) 1.5

(E) 5

$$= \left(\frac{125}{1000} \times \frac{100}{5} \right) \times (0.4 + 0.6)$$

$$= \frac{25}{10} \times 1 = 2.5$$

Q22. $[(0.7)^2 - 0.49] \div 0.07 = ?$

☒ (A) 0

(B) 0.07

(C) 0.49

(D) 7

(E) 1

$$= [(0.7)^2 - 0.49] \div \frac{7}{100}$$

$$= [0.49 - 0.49] \times \frac{100}{7}$$

$$= 0 \times \frac{100}{7} \Rightarrow 0$$

$$\left\{ \begin{array}{l} 7^2 = 49 \\ (0.7)^2 = 0.49 \end{array} \right\}$$

Q23. $\{ (0.08 \times 125) - (0.006 \div 0.0002) \} \times 0.5 = ?$

(A) -5

(B) -20

(C) 10

☒ (D) -10

(E) 0

$$= \{ (0.08 \times 125) - (0.006 \div 0.0002) \} \times 0.5$$

$$= \left\{ \left(\frac{8}{100} \times 125 \right) - \left(\frac{6}{1000} \times \frac{10000}{2} \right) \right\} \times \frac{5}{10}$$

$$\begin{aligned}
 &= \left\{ \left(\frac{8}{100} \times \frac{1000}{8} \right) - \left(6 \times \frac{100}{2} \right) \right\} \times \frac{1}{2} \\
 &= (10 - 30) \times \frac{1}{2} \\
 &= -\frac{20}{2} \Rightarrow -10
 \end{aligned}$$

Q24. $[(1.5 \div 0.3) + (2.4 \div 0.06)] \div (0.5 + 0.25) = ?$

- (A) 45 ☒ (B) 60 (C) 30 (D) 90 (E) 120

$$\begin{aligned}
 &= \left[\left(\frac{15}{10} \div \frac{3}{10} \right) + \left(\frac{24}{10} \div \frac{6}{100} \right) \right] \div \left(\frac{5}{10} + \frac{25}{100} \right) \\
 &= \left[\left(\frac{15}{10} \times \frac{10}{3} \right) + \left(\frac{24}{10} \times \frac{100}{6} \right) \right] \div \frac{75}{100} \\
 &= (5 + 40) \times \frac{100}{75} \Rightarrow 45 \times \frac{4}{3} = 60
 \end{aligned}$$

Q25. $84.25 - 37.18 + 16.93 = ? + 20.50$

- (A) 45.50 (B) 63.50 (C) 41.50 ☒ (D) 43.50 (E) 23.50

$$\begin{aligned}
 84.25 + 16.93 - 37.18 - 20.50 &= x \\
 43 + (0.25 + 0.93 - 0.18 - 0.50) &= x \\
 43 + 0.50 &= x \\
 43.50 &= x
 \end{aligned}$$

Q26. $\sqrt[3]{0.000512} + \sqrt{0.0289} = \sqrt{?}$

- ☒ (A) 0.0625 (B) 0.0256 (C) 0.625 (D) 0.00625 (E) 0.125

$$\begin{aligned}
 &= \sqrt[3]{\frac{512}{1000000}} + \sqrt{\frac{289}{10000}} = \sqrt{x} \\
 &= 0.08 + 0.17 \Rightarrow 0.25 = \sqrt{x} \Rightarrow x = 0.0625
 \end{aligned}$$

Q27. $\sqrt[3]{0.000216} + \sqrt{0.0009} = \sqrt{?}$

- (A) 0.00081 (B) 0.081 ☒ (C) 0.0081 (D) 0.009 (E) 0.0089

$$\sqrt[3]{\frac{216}{1000000}} + \sqrt{\frac{9}{10000}} = \sqrt{x}$$

$$\frac{6}{100} + \frac{3}{100} = \sqrt{x}$$

$$\frac{9}{100} = \sqrt{x}$$

(Squaring both sides)

$$\left(\frac{9}{100}\right)^2 = (\sqrt{x})^2$$

$$x = \frac{81}{10000} = 0.0081$$

Q28. $\frac{(0.12 \times 3.5)}{0.06} + \frac{0.015 \times 400}{0.25} = ?$

(A) 24

☒ (B) 31

(C) 7

(D) 30

(E) 32

$$= \left(\frac{0.12 \times 3.5}{0.06}\right) + \left(\frac{0.015 \times 400}{0.25}\right)$$

$$= \left(\frac{12}{100} \times \frac{35}{10} \times \frac{100}{6}\right) + \left(\frac{15}{1000} \times 400 \times \frac{100}{25}\right)$$

$$= (7) + (24) = 31$$

Q29. $\frac{0.036 \times 0.04}{0.00008} + \frac{0.225 \times 0.8}{0.0036} = ?$

(A) 18

(B) 50

(C) 180

☒ (D) 68

(E) 86

$$\left(\frac{36}{1000} \times \frac{4}{100} \times \frac{100000}{8}\right) + \left(\frac{225}{1000} \times \frac{8}{10} \times \frac{10000}{36}\right)$$

$$= (18) + (50)$$

$$= 68$$

Q30: $1 - \frac{1}{1 + \frac{1}{1.5 + \frac{1}{2.5}}} = ?$

☒ (A) 10/29

(B) 9/29

(C) 11/29

(D) 19/29

(E) None of

these

$$1 - \frac{1}{1 + \frac{1}{\frac{3}{2} + \frac{1}{5/2}}}$$

$\Rightarrow$

$$1 - \frac{1}{1 + \frac{1}{\frac{3}{2} + \frac{2}{5}}}$$

$\Rightarrow$

$$1 - \frac{1}{1 + \frac{10}{19}}$$

$$\Rightarrow 1 - \frac{1}{\frac{29}{19}} \Rightarrow 1 - \frac{19}{29} = \frac{10}{29}$$

$1/2 = 50\%$					
$1/3 = 33.33\%$	$2/3 = 66.67\%$				
$1/4 = 25\%$	$3/4 = 75\%$				
$1/5 = 20\%$	$2/5 = 40\%$	$3/5 = 60\%$	$4/5 = 80\%$		
$1/6 = 16.67\%$	$5/6 = 83.33\%$				
$1/7 = 14.29\%$	$2/7 = 28.57\%$	$3/7 = 42.86\%$	$4/7 = 57.14\%$	$5/7 = 71.43\%$	$6/7 = 85.71\%$
$1/8 = 12.5\%$	$3/8 = 37.5\%$	$5/8 = 62.5\%$	$7/8 = 87.5\%$		
$1/9 = 11.11\%$	$2/9 = 22.22\%$	$4/9 = 44.44\%$	$5/9 = 55.56\%$	$7/9 = 77.78\%$	$8/9 = 88.89\%$
$1/10 = 10\%$					
$1/11 = 9.09\%$	$2/11 = 18.18\%$	$3/11 = 27.27\%$	$4/11 = 36.36\%$	$5/11 = 45.45\%$	$6/11 = 54.55\%$
	$7/11 = 63.63\%$	$8/11 = 72.72\%$	$9/11 = 81.81\%$	$10/11 = 90.90\%$	
$1/12 = 8.33\%$	$5/12 = 41.67\%$	$11/12 = 91.67\%$			
$1/13 = 7.69\%$	$2/13 = 15.38\%$				
$1/14 = 7.14\%$	$3/14 = 21.43\%$	$11/14 = 78.57\%$	$13/14 = 92.86\%$		
$1/15 = 6.67\%$	$2/15 = 13.33\%$	$4/15 = 26.67\%$	$7/15 = 46.67\%$	$8/15 = 53.33\%$	$11/15 = 73.33\%$
	$13/15 = 86.67\%$	$14/15 = 93.33\%$			
$1/16 = 6.25\%$	$3/16 = 18.75\%$	$5/16 = 31.25\%$	$7/16 = 43.75\%$	$9/16 = 56.25\%$	$11/16 = 68.75\%$
	$13/16 = 81.25\%$	$15/16 = 93.75\%$			
$1/17 = 5.88\%$					
$1/18 = 5.56\%$					
$1/19 = 5.26\%$					
$1/20 = 5\%$	$3/20 = 15\%$	$7/20 = 35\%$	$11/20 = 55\%$	$13/20 = 65\%$	
$1/21 = 4.76\%$					
$1/22 = 4.55\%$					
$1/23 = 4.35\%$					
$1/24 = 4.17\%$					
$1/25 = 4\%$	$2/25 = 8\%$	$3/25 = 12\%$			
$1/30 = 3.33\%$	$7/30 = 23.33\%$				

• Percent Meaning

$$x\% = \frac{x}{100}$$

Example, $35\% = \frac{35}{100} = \frac{7}{20}$

$$125\% = \frac{125}{100} = \frac{5}{4}$$

• "Of" Rule

$$x\% \text{ of } N = \frac{x}{100} \times N$$

Example, $25\% \text{ of } 360 = \frac{25}{100} \times 360 = \frac{1}{4} \times 360 = 90$

• 1% Method

$$1\% \text{ of } N = N \times \frac{1}{100} = \frac{N}{100}$$

$$5\% \text{ of } N = N \times \frac{5}{100} = \frac{N}{20}$$

$$2\% \text{ of } N = N \times \frac{2}{100} = \frac{N}{50}$$

$$0.5\% \text{ of } N = N \times \frac{5}{1000} = \frac{N}{200}$$

$$0.25\% \text{ of } N = N \times \frac{25}{100} \times \frac{1}{100} = \frac{N}{400}$$

Example,

$$0.5\% \text{ of } 2400 = \frac{2400}{200} = 12$$

• 99% Method

$$99\% \text{ of } N = (100-1)\% \text{ of } N = N - 1\% \text{ of } N$$

$$98\% \text{ of } N = (100-2)\% \text{ of } N = N - 2\% \text{ of } N$$

$$95\% \text{ of } N = (100-5)\% \text{ of } N = N - 5\% \text{ of } N$$

Example,

$$99\% \text{ of } 700 = 700 - 7 = 693$$

- Reciprocal of %
(Division by %)

$$N \div P\% = \frac{N}{P/100} = N \times \frac{100}{P}$$

Example, $200 \div 25\% = 200 \div \frac{1}{4}$
 $= 200 \times 4 = 800$

- Golden Identity-

$$a\% \text{ of } b = b\% \text{ of } a$$

Example,

$$48\% \text{ of } 25 = 25\% \text{ of } 48 = 12$$

$$64\% \text{ of } 75 = 75\% \text{ of } 64 = \frac{3}{4} \times 64 = 48$$

$$128\% \text{ of } 50 = 50\% \text{ of } 128 = \frac{1}{2} \times 128 = 64$$

$$84\% \text{ of } 50 = 50\% \text{ of } 84 = \frac{1}{2} \times 84 = 42$$

$$36\% \text{ of } 25 = 25\% \text{ of } 36 = \frac{1}{4} \times 36 = 9$$

Percentage

Q1: 50% of 840 = ?

(A) 380

(B) 400

☒ (C) 420

(D) 48

(E) 440

$$\frac{1}{2} \times 840 = 420$$

Q2: 16.67% of 960 = ?

☒ (A) 160

(B) 200

(C) 240

(D) 260

(E) 280

$$= 16.67\% \times 960$$

$$= \frac{1}{6} \times 960 = 160$$

Q3: 12.5% of 640 = ?

(A) 60

(B) 70

☒ (C) 80

(D) 90

(E) 100

$$= 12.5\% \times 640$$

$$= \frac{1}{8} \times 640 = 80$$

Q4: $16\frac{2}{3}\%$ of 300 = ?

- (A) 40 ☒ (B) 50 (C) 60 (D) 75 (E) None of these

$$16.67\% \text{ of } 300$$

$$= \frac{1}{6} \times 300 = 50$$

Q5: 6.25% of 800 = ?

- (A) 30 (B) 32 (C) 40 ☒ (D) 50 (E) 60

$$= \frac{1}{16} \text{ of } 800$$

$$= \frac{1}{16} \times 800 = 50$$

Q6: 0.5% of 2400 = ?

- (A) 10 ☒ (B) 12 (C) 14 (D) 16 (E) 18

$$= \frac{1}{200} \times 2400$$

$$= 12$$

$$\left\{ \begin{array}{l} 50\% = \frac{1}{2} \\ 5\% = \frac{1}{20} \\ 0.5\% = \frac{1}{200} \end{array} \right\}$$

Q7: 70% of 160 × ? = 112% of 90

(A) 0.80

☒ (B) 0.90

(C) 1

(D) 1.10

(E) 1.20

$$\left(\frac{7}{10} \times 160\right) \times x = \frac{112}{100} \times 90$$

$$x = \frac{112}{100} \times 90 \times \frac{1}{16 \times 7} = \frac{9}{10} = 0.9$$

Q8: 30% of 400 + 15% of 200 = ?

(A) 140

☒ (B) 150

(C) 160

(D) 170

(E) 180

$$= \left(\frac{3}{10} \times 400\right) + \left(\frac{3}{20} \times 200\right)$$

$$= 120 + 30$$

$$= 150$$

Q9: 75% of 360 - 12.5% of 640 = ?

(A) 150

(B) 160

(C) 170

(D) 180

☒ (E) 190

$$\left(\frac{3}{4} \times 360\right) - \left(\frac{1}{8} \times 640\right)$$

$$\Rightarrow 270 - 80 \Rightarrow 190$$

Q10: 33.33% of 270 + 5.88% of 340 = ?

(A) 100

☒ (B) 110

(C) 120

(D) 130

(E) None of

these

$$= \left(\frac{1}{3} \times 270\right) + \left(\frac{1}{17} \times 340\right)$$

$$= 90 + 20 = 110$$

Q11: 62.5% of 160 - 37.5% of 80 = ?

- (A) 40 (B) 50 (C) 60 ☒ (D) 70 (E) None of

these

$$\begin{aligned}
 &= \left(\frac{5}{8} \times 160 \right) - \left(\frac{3}{8} \times 80 \right) \\
 &= 100 - 30 \\
 &= 70
 \end{aligned}$$

Q12: 1.25% of 32000 - 80% of 250 = ?

- (A) 150 (B) 180 ☒ (C) 200 (D) 220 (E) 240

$$\begin{aligned}
 &= \left(\frac{1}{80} \times 32000 \right) - \left(\frac{8}{10} \times 250 \right) \\
 &= 400 - 200 \\
 &= 200
 \end{aligned}$$

Q13: 480 ÷ 12.5% = ?

- (A) 60 (B) 240 (C) 2400 ☒ (D) 3840 (E) None of these

$$\begin{aligned}
 \frac{480}{\frac{1}{8}} &= 480 \times 8 \\
 &= 3840
 \end{aligned}$$

Q14: $25\% \text{ of } 400 + 120 \div 30\% = ?$

- (A) 400 ☒ (B) 500 (C) 600 (D) 720 (E) None of

these

$$= \left(\frac{1}{4} \times 400 \right) + \left(\frac{120}{\frac{30}{100}} \right)$$

$$= 100 + 400$$

$$= 500$$

Q15: $2500 \times 0.4\% = ?$

- (A) 8 ☒ (B) 10 (C) 12 (D) 14 (E) 16

$$2500 \times \frac{1}{250} = 10$$

$$\left\{ \begin{array}{l} 4\% = \frac{1}{25} \\ 0.4\% = \frac{1}{250} \end{array} \right\}$$

Q16: $(? + 20)\% \text{ of } 500 = 350$

- (A) 45 ☒ (B) 50 (C) 55 (D) 60 (E) 65

$$\frac{(x + 20)}{100} \times 500 = 350$$

$$x + 20 = 70$$

$$x = 50$$

Q17: $(? - 15)\%$ of 800 = 520% of 100

(A) 50

(B) 75

☒ (C) 80

(D) 90

(E) 70

$$\frac{(x-15)}{100} \times 800 = 520$$

$$x - 15 = 65$$

$$x = 80$$

Q18: 24% of 1200 $\div$ 144% of 40 = ?

(A) 1

(B) 2

(C) 3

(D) 4

☒ (E) 5

$$\left(\frac{24}{100} \times 1200 \right) \div \left(\frac{144}{100} \times 40 \right)$$

$$2 \times \frac{288}{144} \times \frac{10}{4} = 5$$

Q19: $[(25\% \text{ of } 800) - (12.5\% \text{ of } 640)] \div (10\% \text{ of } 200) = ?$

(A) 5

☒ (B) 6

(C) 4

(D) 8

(E) None of

these

$$\left[\left(\frac{1}{4} \times 800 \right) - \left(\frac{1}{8} \times 640 \right) \right] \div \left(\frac{1}{10} \times 200 \right)$$

$$= (200 - 80) \times \frac{1}{20}$$

$$= \frac{120}{20} = 6$$

Q20: 0.5% of 0.2% of 500000 = ?

(A) 2

☒ (B) 5

(C) 1

(D) 10

(E) 12

$$\frac{1}{200} \times \frac{1}{500} \times 500000$$

$$= 5$$

$$\left\{ \begin{array}{l} 20\% = \frac{1}{5} \\ 2\% = \frac{1}{50} \\ 0.2\% = \frac{1}{500} \end{array} \right\} \left\{ \begin{array}{l} 50\% = \frac{1}{2} \\ 5\% = \frac{1}{20} \\ 0.5\% = \frac{1}{200} \end{array} \right\}$$

Q21: $(1 \div 12.5\%) + (1 \div 6.25\%) = ?$

(A) 2.40

☒ (B) 24

(C) 240

(D) 480

(E) None of

these

$$\left(\frac{1}{\frac{1}{8}} \right) + \left(\frac{1}{\frac{1}{16}} \right)$$

$$= 8 + 16$$

$$= 24$$

Q22: $\left\{ \frac{[83.33\% \text{ of } 720]}{[1.667\% \text{ of } 3600]} \right\} + 5 = ?$

(A) 12

(B) 13

(C) 14

☒ (D) 15

(E) 16

$$= \left\{ \frac{\frac{5}{6} \times 720}{\frac{1}{60} \times 3600} \right\} + 5$$

$$= \left(\frac{120 \times 5}{60} \right) + 5$$

$$= 10 + 5$$

$$= 15$$

Q23: $\frac{480}{12.5\%} - \frac{360}{16.67\%} + \frac{200}{20\%} = ?$

- (A) 2480 (B) 2580 (C) 2680 (D) 2780 (E) 2880

$$\begin{aligned} & \left(\frac{480}{\frac{1}{8}} \right) - \left(\frac{360}{\frac{1}{6}} \right) + \left(\frac{200}{\frac{1}{5}} \right) \\ &= (480 \times 8) - (360 \times 6) + (200 \times 5) \\ &= 3840 - 2160 + 1000 \Rightarrow 2680 \end{aligned}$$

Q24: $[0.833\% \text{ of } 9600 + 41.67\% \text{ of } 600] \times (12.5\% \text{ of } 80) = ?$

- (A) 3000 (B) 3100 (C) 3200 (D) 3300 (E) 3400

$$\begin{aligned} &= \left[\left(\frac{1}{120} \times 9600 \right) + \left(\frac{5}{12} \times 600 \right) \right] \times \left(\frac{1}{8} \times 80 \right) \\ &= (80 + 250) \times 10 \\ &= 330 \times 10 \Rightarrow 3300 \end{aligned}$$

Q25: $\frac{[71.4\% \text{ of } 140 + 21.43\% \text{ of } 1400]}{28.57\% \text{ of } 700} = ?$

- (A) 1 (B) 1.5 (C) 2 (D) 13 (E) 6.5

$$\begin{aligned} &= \frac{\left(\frac{5}{7} \times 140 \right) + \left(\frac{3}{14} \times 1400 \right)}{\frac{2}{7} \times 700} \Rightarrow \frac{100 + 300}{200} \Rightarrow \frac{400}{200} \Rightarrow 2 \end{aligned}$$

Q26: $[0.84\% \text{ of } 50 + 25\% \text{ of } 0.64] \times 1000 = ?$

- (A) 540 (B) 580 (C) 620 (D) 640 (E) 680

$$[0.50\% \text{ of } 84 + 25\% \text{ of } 0.64] \times 1000$$

$$\left[\frac{1}{200} \times 84 + \frac{1}{4} \times 0.64 \right] \times 1000$$

$$(0.42 + 0.16) \times 1000$$

$$= 0.58 \times 1000 \Rightarrow 580$$

{ Golden Identity }

Q27: $\left(\frac{7}{5} - 50\% \text{ of } \frac{8}{5}\right) = ?$

(A) 40%

(B) 50%

☒ (C) 60%

(D) 70%

(E) 80%

$$= \left(\frac{7}{5}\right) - \left(\frac{1}{2} \times \frac{8}{5}\right)$$

$$= \frac{7}{5} - \frac{4}{5} = \frac{3}{5} = 60\%$$

Q28: 5.88% of $\left(\frac{3}{5} + \frac{1}{4}\right) = ?$

(A) 4%

☒ (B) 5%

(C) 6%

(D) 7%

(E) 8%

$$= \frac{1}{17} \times \left(\frac{3}{5} + \frac{1}{4}\right)$$

$$= \frac{1}{17} \times \left(\frac{12+5}{20}\right) \Rightarrow \frac{1}{17} \times \frac{17}{20} \Rightarrow \frac{1}{20} \Rightarrow 5\%$$

Q29: 4.54% of $\left(\frac{3}{5} + \left(\frac{7}{5} \times \frac{1}{5}\right)\right) = ?$

(A) 3%

☒ (B) 4%

(C) 5%

(D) 6%

(E) 7%

$$= \frac{1}{22} \times \left(\frac{3}{5} + \frac{7}{25}\right)$$

$$= \frac{1}{22} \times \left(\frac{15+7}{25}\right)$$

$$= \frac{1}{22} \times \frac{22}{25} \Rightarrow \frac{1}{25} \Rightarrow 4\%$$

Q30: $\left(\left(\frac{4}{13} \times 4\right) + \frac{5}{13}\right)$ of $66.67\% - 100\% = ?$

(A) 5%

(B) 1%

(C) 7.14%

(D) 5.88%

(E) None of

these

$$= \left[\left(\frac{4}{13} \times 4 \right) + \frac{5}{13} \right] \times \left(\frac{2}{3} \right) - 100\%$$

$$= \left[\frac{16}{13} + \frac{5}{13} \right] \times \frac{2}{3} - 100\%$$

$$= \left(\frac{21}{13} \times \frac{2}{3} \right) - \frac{100}{100}$$

$$= \frac{14}{13} - \frac{1}{1} = \frac{1}{13}$$